https://learn.genetics.utah.edu/content/labs/extraction/

http://www.explorecuriocity.org/portals/2/themes/biotechnology/DNA-Extraction-Backgrounder.pdf

1. Give three examples of why we might need to extract DNA from humans. (3)

Why would a blood sample not be as ideal for DNA extraction? (2)

Red blood cells don't contain a nucleus, only white, less concentration

*What type of tissue are buccal (cheek) cells? What makes them suited to the purpose of DNA extraction? (2)
Squamous epithelial
Shed often and easily

What are the 4 steps in DNA extraction? (4)
Collect cell sample
Release DNA through lysis
Isolate DNA
Precipitate

What are the two main ingredients in the lysis solution and what do they do? (4)
Detergent – break down cell and organelle membranes
Enzyme (Proteinase K) – breaks down proteins

*Why is the sample put in a warm water bath after the lysis solution is added? (2)
Optimum temperature for enzyme activity

Describe the 2 methods of physical separation involved in separating the DNA from the proteins and other debris. (2)
Centrifugation – causes solids to come out of suspension
Decanting - using pipette to remove solution

What does adding the isopropyl alcohol do and why? (2)
Causes DNA to precipitate out of solution because it is insoluble in alcohol

*Sometimes the DNA can become contaminated with other DNA not from the target organism. List 2 possible contaminants and describe where in the process they might occur. (4)
Bacterial DNA – from mouth
DNA from lab technician – handling of equipment

*For the strawberry lab, research why strawberries are often used for DNA extractions.
Define polyploidy (1): more than one set of chromosomes/many sets

Explain why strawberries are used (2): Many sets of chromosomes = larger volume of DNA = more easy to observe

In the Strawberry DNA extraction we used mechanical and chemical lysis. Describe both of these processes in the lab. (2)
Mechanical = squashing strawberries with hands, Chemical = detergent

Propose a reason why the human sample did not use mechanical lysis but the strawberry lab does (1)
Plant has cell walls = need to break down more

What step/s in the virtual lab are analogous to the filtering of the strawberry pulp? (2)
Centrifuging, decanting

What is left behind in the filtrate? (2)
Lipids from cell membranes, proteins

	DNA concentration (ng/µl)			
	Trial 1	Trial 2	Trial 3	Average
Raspberry	130	160	134	
Pear	24	25	23	
Strawberry	100	95	105	
Grape	28	41	31	

	Amount of DNA extracted (grams)			
	Trial 1	Trial 2	Trial 3	Average
Banana	5.6	6.2	11.1	
Kiwifruit	7.3	7.9	8.5	
Strawberry	11.4	12.3	10.7	

Create a graph for this data (5)
Title relates dependent and independent variables
Appropriate labels and units
Type of graph appropriate
Appropriate size and scale
Calculating averages correctly and graphing the appropriate data (removing outlier, only graphing averages)

Write a hypothesis which the researcher may have written. (2)

What was the independent variable? (1)

What was the dependent variable? (1)
List 5 things that would have been controlled in this experiment. (5)
What might you infer from this data regarding the content of DNA extracted and polyploidy? (1)

A person cannot see a single cotton thread 100 feet away, but if you wound thousands of threads together into a rope, it would be visible at some distance. How is this statement an analogy to our DNA extraction?

Mark /50?